

Session 4.1

Aiyagari, Solved Twice: A Fixed Point, Then a Law of Motion

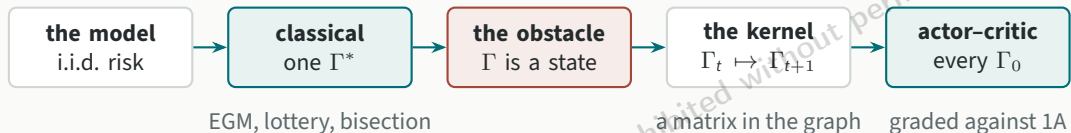
The classical solve returns one stationary distribution. An actor-critic solve with the distribution as its state returns how every distribution evolves



Zhigang Feng

The pages marked PRELIMINARY are based on preliminary research by Zhigang Feng, Jiequn Han, Thomas J. Sargent, and Shenghao Zhu (September 2026). Do not circulate without the authors' permission.

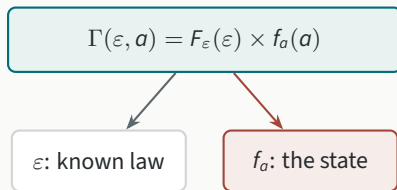
From a Fixed Point to a Dynamical System



- ▶ One economy throughout: Aiyagari with productivity redrawn each period (i.i.d.), no government, no aggregate risk.
- ▶ **Stage 1A**, classical: the stationary distribution Γ^* and the prices that support it.
- ▶ **Stage 1B**, networks that read the whole distribution: the map $\Gamma_{t+1} = \Phi(\Gamma_t)$, so the path from any start.
- ▶ **Stage 1A is the benchmark Stage 1B is graded against: same calibration, same quadrature.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Aiyagari, *QJE* (1994).

The Economy



ε is redrawn every period, independently of a , so today's assets tell nothing about today's ε

- ▶ $\log \varepsilon \sim N(0, 0.2^2)$, i.i.d.;
 $c + a' = (1 + r)a + w\varepsilon l, a' \geq 0$.
- ▶ Utility $(c - \psi_l \frac{l^{1+\theta_l}}{1+\theta_l})^{1-\sigma} / (1 - \sigma)$: no wealth effect on hours, so $l(\varepsilon) = (w\varepsilon / \psi_l)^{1/\theta_l}$.
- ▶ $Y = K^\alpha L^{1-\alpha}$; r (net of δ) and w from marginal products; $K = \int a d\Gamma$.
- ▶ $\Gamma_{t+1} = \Phi(\Gamma_t)$, and with i.i.d. ε the asset marginal f_a is the whole state.
- ▶ **Prices depend on Γ , so the saving rule is $g(\varepsilon, a; \Gamma)$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\beta = 0.95, \sigma = 2, \alpha = 0.34, \delta = 0.06, \theta_l = 0.32, \psi_l = 1.5$.

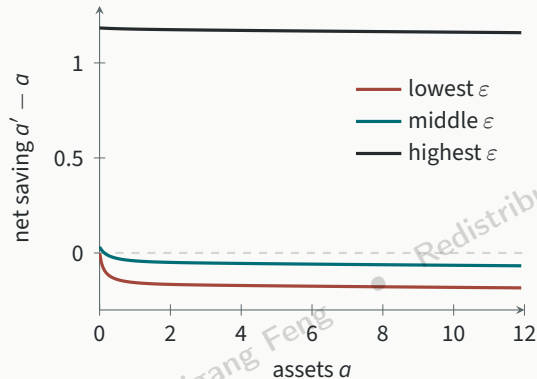
Session 4.1

Stage 1A: The Classical Solve

A household, a population and a price, in the benchmark every later number is graded against



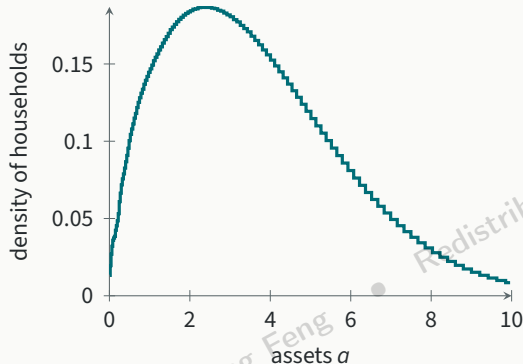
The Household: the Endogenous Grid Method



- ▶ Grid on tomorrow's a' ; invert the Euler equation for c ; today's a from the budget.
- ▶ Below the first recovered a the limit binds: $a' = 0$. A comparison, not a solver.
- ▶ 1500 asset nodes; 5 Gauss–Hermite nodes for ε , as the network uses later.
- ▶ **The lucky save, the unlucky dissave. The economics is the gap between these lines and zero.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Carroll, *Economics Letters* (2006).

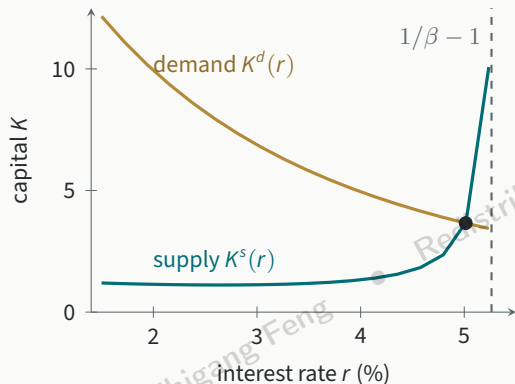
Young's Lottery



- ▶ Mass whose a' lands between nodes a_j, a_{j+1} is split: $\omega = \frac{a' - a_j}{a_{j+1} - a_j}$ up, $1 - \omega$ down.
- ▶ Mass is conserved and the mean of a' is exact: next period's K is exact, given the policy.
- ▶ Stationary λ : one dense solve of $(P_g^\top - I)\lambda = 0, \sum \lambda = 1$.
- ▶ Result: 0.009% at $a = 0$, Gini 0.338.
- ▶ **A histogram plus this lottery is the law of motion, as a matrix.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Young, *JEDC* (2010).

The Price: Bisection Inside a Proven Bracket



- ▶ Each supply point is one household solve and one distribution solve.
- ▶ Supply explodes as $r \rightarrow 1/\beta - 1$: the bracket's top. Prove a sign change before bisectioning.
- ▶ $r^* = 5.014\%$, $K^* = 3.6595$, $K/Y = 3.09$; about 30 CPU-seconds.
- ▶ $\beta(1 + r^*) = 0.99763$. **That number, so close to one, returns in every section that follows.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Aiyagari (1994).

Is the Benchmark Right?

Check	Measured	What it rules out
a second algorithm: value iteration	gap 0.0489 at 400 nodes, 0.0251 at 800: halves	an error in the EGM code
grid refinement, 800 against 1500 nodes	K moves by 0.003%	a grid that sets the answer
a panel of 200 000 households	mean drifts 1.4×10^{-5} in 1000 periods	an error in the lottery
Euler equation, with the limit market clearing at r^*	● worst interior error 5.9×10^{-5} $ K^s - K^d = 3.6 \times 10^{-6}$	a policy that is not optimal an early stop

- **Five checks, each aimed at a different piece. Without them there is nothing to grade a network against.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A.

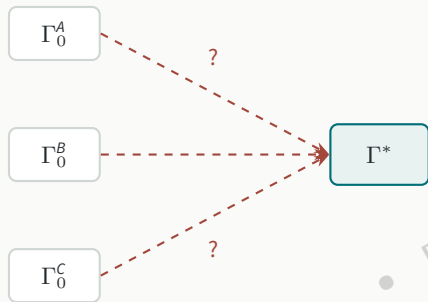
Session 4.1

What the Classical Solve Cannot Answer

One fixed point is not a law of motion, and the distribution is now a state



One Fixed Point, Not a Law of Motion



each dashed path is a separate problem: a sequence of prices, solved backward and forward

- ▶ Stage 1A says where the economy rests, not how it gets there.
- ▶ A transition from one Γ_0 is a price path and a shooting loop (Deck 4.3). A second Γ_0 is a second loop.
- ▶ Welfare along the way, or a policy that reacts to Γ , needs the answer at every Γ .
- ▶ **The unknown changes from a number, r^* , to a function of a distribution: $g(\varepsilon, a; \Gamma)$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Deck 4.3, *The Unknown Is a Path of Prices*.

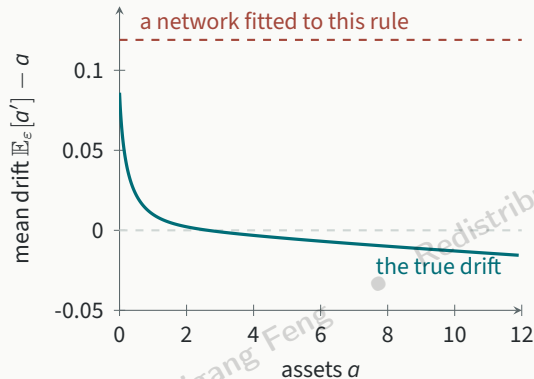
The Distribution Is a State

Route	What the network reads	Its weakness here
a simulated panel	every household's (ε, a) , in some order	noise, and an ordering to forget
generalised moments	learned averages of the panel	the moments must be chosen or learned
a histogram	the masses in N fixed bins	N inputs; the grid must be chosen

- ▶ Each route turns an infinite-dimensional Γ into a finite vector. Deck 4.2 takes the first two; here, $N = 140$ bins.
- ▶ **A histogram has no ordering to forget, and it can be moved forward exactly, by a matrix.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Maliar et al. (2021); Azinovic et al. (2022); Han, Yang & E (2026).

A Near Unit Root Makes Small Errors Large



- ▶ The true rule's mean drift stays within 0.022 of zero for $0.5 \leq a \leq 10$.
- ▶ With $\beta(1+r) = 0.99763$, a one-signed bias in it is amplified about $1/(1 - \beta R) \approx 420$ times.
- ▶ Measured: a network outputting the level of a' , fitted to the exact rule, kept a drift of +0.119; its K was 15.3, not 3.6595.
- ▶ **So the network outputs a change in assets.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A and the network initialisation.

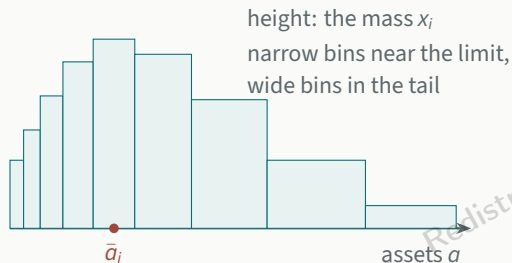
Session 4.1

Characterising the Evolution of Γ

A histogram, a quadrature, and a transition matrix rebuilt inside the computational graph



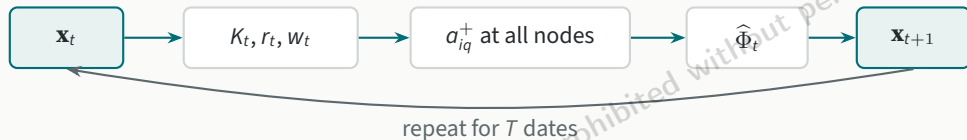
A Histogram and a Quadrature



- ▶ $N = 140$ bins on $[0, 16]$ with midpoints $\bar{a}_i$; the state is $\mathbf{x}$, $\sum_i x_i = 1$, $K = \sum_i x_i \bar{a}_i$.
- ▶ ε : $Q = 5$ Gauss-Hermite nodes ε_q , weights p_q . An expectation is a five-term sum.
- ▶ Grid chosen once from the benchmark: no bin above 1.6% of households, nothing off the top.
- ▶ **The discretisation is part of the model. Fix it for every method compared.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

The Law of Motion, Rebuilt at Every Date



From one distribution $\mathbf{x}_0$, for $t = 0, \dots, T-1$

- 1 $K_t = \sum_i x_{t,i} \bar{a}_i$; r_t, w_t from the firm, hours in closed form: no price loop
- 2 saving $a_{iq}^+ = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}_t)$ at every node; consumption from the budget
- 3 build the $N \times N$ matrix $\hat{\Phi}_t$ and set $\mathbf{x}_{t+1} = \mathbf{x}_t \hat{\Phi}_t$

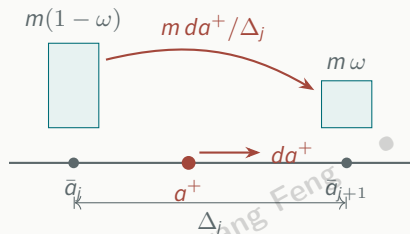
► $\hat{\Phi}_t$ depends on $\mathbf{x}_t$ and on the network, so it is rebuilt at every date of every rollout: computed, never simulated, inside the graph.

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Building $\hat{\Phi}$, and Why This Kernel

For every source midpoint $\bar{a}_i$ and node ε_q

- 1 evaluate the saving network once: $a^+ = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x})$
- 2 find the bracket $\bar{a}_j \leq a^+ \leq \bar{a}_{j+1}$, and the weight $\omega = (a^+ - \bar{a}_j) / (\bar{a}_{j+1} - \bar{a}_j)$
- 3 add $p_q(1 - \omega)$ to $\hat{\Phi}_{ij}$ and $p_q \omega$ to $\hat{\Phi}_{i,j+1}$



- **Exact in the mean:** the only split between neighbours with mean a^+ .
- **Mass by construction:** each row of $\hat{\Phi}$ sums to one.
- **Differentiable:** da^+ moves mass $m da^+ / \Delta_j$ between neighbours, and autodiff chains it across dates.
- **Stationary K within 0.06% of the exact operator's; an inversion in ε was off by 17%.**

Session 4.1

Stage 1B: An Actor–Critic Solve

The saving rule is the actor, the household's value is the critic, and both read the histogram



The Problem the Two Networks Solve

- At a distribution $\mathbf{x}$, each household solves, taking $\mathbf{x}$, prices and $\mathbf{x}_+$ as given,

$$V(\varepsilon, a; \mathbf{x}) = \max_{0 \leq a' \leq \bar{a}} \left\{ u\left(c - \psi_l \frac{l^{1+\theta_l}}{1+\theta_l}\right) + \beta \sum_{q'} p_{q'} V(\varepsilon_{q'}, a'; \mathbf{x}_+) \right\}, \quad u(y) = \frac{y^{1-\sigma}}{1-\sigma},$$

with $c = (1 + r(\mathbf{x})) a + w(\mathbf{x}) \varepsilon l(\varepsilon; \mathbf{x}) - a'$ and $\mathbf{x}_+ = \mathbf{x} \hat{\Phi}(\mathbf{x}; \theta)$.

- **Actor:** $a' = g_\theta(\varepsilon, a; \mathbf{x}) = \min\{\max\{\tilde{a}, 0\}, \bar{a}\}$, $\tilde{a} = a + D(2s_\theta - 1)$, $s_\theta \in (0, 1)$, $D = 1.3$.
- **Critic:** $V_\psi(\varepsilon, a; \mathbf{x}) \approx V$, and a slowly moving copy $V_{\psi-}$, the **target network**.
- Each: two hidden layers of 256 rectified-linear units, reading (ε, a) and the 140 masses.
- **No government, so no one else's problem: prices are marginal products at $\mathbf{x}$, and the Euler equation is the household's own first-order condition.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

The Critic Step: Fit the Bellman Equation

- For a batch of distributions $\{\mathbf{x}^e\}$ and every node $(\bar{a}_i, \varepsilon_q)$, with $a_{iq}^+ = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}^e)$ from the current actor:

$$\min_{\psi} \mathcal{L}_V(\psi) = \text{mean}_{e,i,q} \left[V_{\psi}(\varepsilon_q, \bar{a}_i; \mathbf{x}^e) - y_{iq}^e \right]^2, \quad y_{iq}^e = U_{iq}^e + \beta \sum_{q'} p_{q'} V_{\psi^-}(\varepsilon_{q'}, a_{iq}^+; \mathbf{x}_+^e).$$

- U_{iq}^e is flow utility at a_{iq}^+ . The target y is held fixed while ψ moves: no gradient flows into it.
- One Adam step, learning rate 3×10^{-3} ; then $\psi^- \leftarrow (1 - 0.02) \psi^- + 0.02 \psi$.
- The expectation over ε' is an exact five-node sum: the target carries no sampling noise.
- **8 such steps per iteration; before training starts, 6000 with the actor frozen.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

The Actor Step: Raise the Right-Hand Side

- ▶ With the critic fixed, the actor maximises the Bellman right-hand side at every node:

$$\max_{\theta} J(\theta) = \text{mean}_{e,i,q} \left[U(a_{iq}^+(\theta)) + \beta C_{iq}(a_{iq}^+(\theta)) - \kappa B_{iq}(\theta) \right],$$

where $C_{iq}(a') = \sum_{q'} p_{q'} V_{\psi^-}(\varepsilon_{q'}, a'; \mathbf{x}_+^e)$ is the continuation and $\kappa = 1$.

- ▶ Barrier $B = [\max(-c, 0) + \max(-c + \psi_l \frac{l^1 + \theta_l}{1 + \theta_l}, 0)] / \eta_c$, $\eta_c = 1 \times 10^{-4}$: zero where consumption is feasible.
- ▶ The slope of C is replaced: its *value* comes from the critic, its *derivative* from the envelope condition, $\partial C / \partial a' \equiv \sum_{q'} p_{q'} (1 + r(\mathbf{x}_+)) u'(c_{+,q'})$, with c_+ from the current actor at $(\varepsilon_{q'}, a'; \mathbf{x}_+)$.
- ▶ So $\partial J / \partial a^+ = -u'(c) + \beta \mathbb{E}[(1 + r_+) u'(c_+)]$, the Euler residual.
- ▶ **Why: at $\beta R = 0.99763$ the two terms nearly cancel; a fitted critic's slope is too noisy to resolve it.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

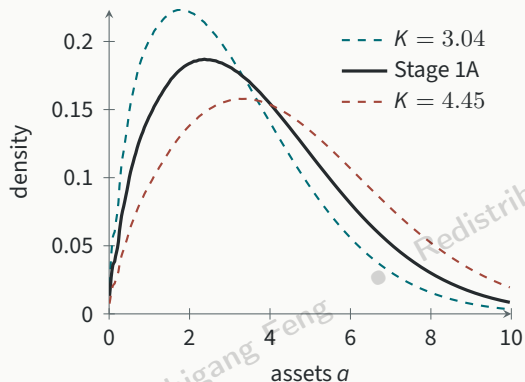
The Actor's Full Loss: One Economic Term, Two Solver Terms

$$\mathcal{L}_g(\theta) = -J(\theta) + \lambda_K \frac{1}{T} \sum_{t=1}^T (K_t(\theta) - K^*)^2 + \lambda_E \text{mean}_e \sum_{i,q} x_i^e p_q |\text{FB}(\vartheta_{1,iq}, \vartheta_{2,iq})|$$

- **Rollout:** $\mathbf{x}_0 = \mathbf{x}^*$, $\mathbf{x}_{t+1} = \mathbf{x}_t \hat{\Phi}(\mathbf{x}_t; \theta)$, $K_t = \sum_i x_{t,i} \bar{a}_i$; $T = 100$, $\lambda_K = 1000$. The gradient passes through all T matrices. The target is Stage 1A's K^* : **agreement on K is imposed.**
- **Euler with the limit:** $\text{FB}(\vartheta_1, \vartheta_2) = \vartheta_1 + \vartheta_2 - \sqrt{\vartheta_1^2 + \vartheta_2^2 + \eta^2}$, with $\vartheta_1 = [u'(c) - \beta \mathbb{E}(1 + r_+) u'(c_+)] / u'(c)$ and $\vartheta_2 = \tilde{a} / 0.1$; $\eta = 0.01$, $\lambda_E = 10$.
- $\text{FB} = 0$ exactly when $\vartheta_1 \geq 0$, $\vartheta_2 \geq 0$ and one is zero: the Euler equation holds, or the limit binds. Weighted by where households are.
- **2 Adam steps per iteration, rate 5×10^{-5} . At βR near one, the rollout, not a residual, pins K .**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B; Maliar, Maliar & Winant (2021).

Training Data: Whole Distributions



- ▶ A training example is a whole distribution $\mathbf{x}^e$, never a household.
- ▶ $x_i^e \propto x_i^* e^{\kappa \bar{a}_i}$, κ uniform on ± 0.14 : a tilt of the benchmark.
- ▶ K spans 3.04 to 4.45. The band sets where losses are evaluated, nothing else.
- ▶ **At each $\mathbf{x}^e$ every node $(\bar{a}_i, \varepsilon_q)$ is evaluated, weighted $x_i p_q$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

The Algorithm

Stage 1B: the household's actor-critic, no government

- 1 initialise the actor by regression on EGM rules at constant prices, one per sampled K
 - 2 pre-fit the critic to that actor, 6000 steps, so the actor cannot exploit a bad critic
 - 3 draw 128 distributions from the tilted band
 - 4 8 critic steps on $\mathcal{L}_V$, each followed by $\psi^- \leftarrow (1 - 0.02)\psi^- + 0.02\psi$
 - 5 2 actor steps on $\mathcal{L}_g$: envelope gradient, rollout penalty, Euler term
 - 6 repeat 4000 times; at intervals, find the actor's own stationary $\mathbf{x}$ and score it
- ▶ Step 1 means the training loop starts at a correct answer. The test is whether it keeps it while learning how the rule depends on $\mathbf{x}$.
 - ▶ **The final run: about 2.9 GPU-hours, against 30 CPU-seconds for Stage 1A.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

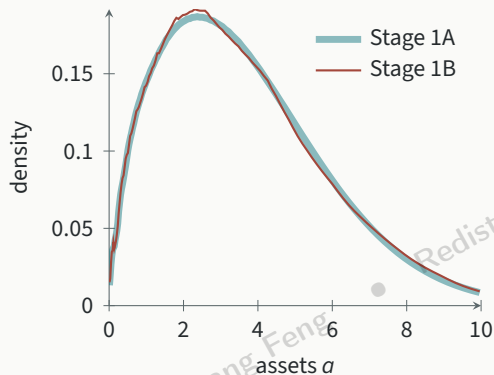
Session 4.1

What It Returns, and Where It Stops

The network against the benchmark, the distribution in motion, and why persistence changes the problem



Stage 1B Against Stage 1A

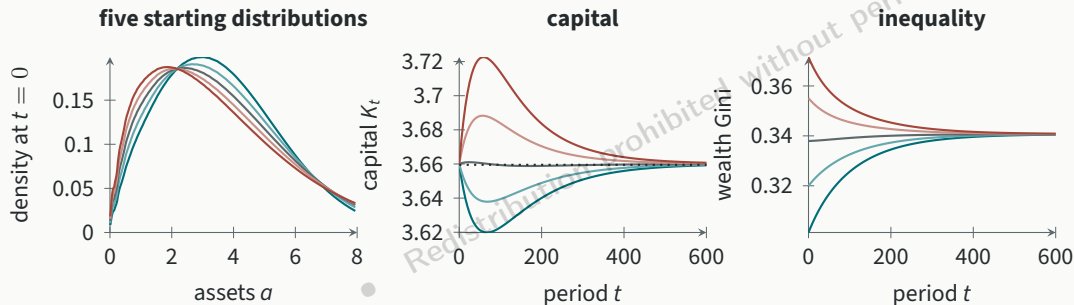


	1A	1B	gap
capital K	3.6595	3.6600	0.01%
wealth Gini	0.3381	0.3407	0.8%
mass below $a = 0.6$	0.0415	0.0419	0.9%
L_1 distance			0.024

- K is held by the penalty. The Gini, the mass near the limit and the whole density are not targeted, and agree to under 1%.
- Euler error, mass-weighted: 0.16%; over all nodes, 0.48%.

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B. The 1A density is integrated onto the network's 140 bins.

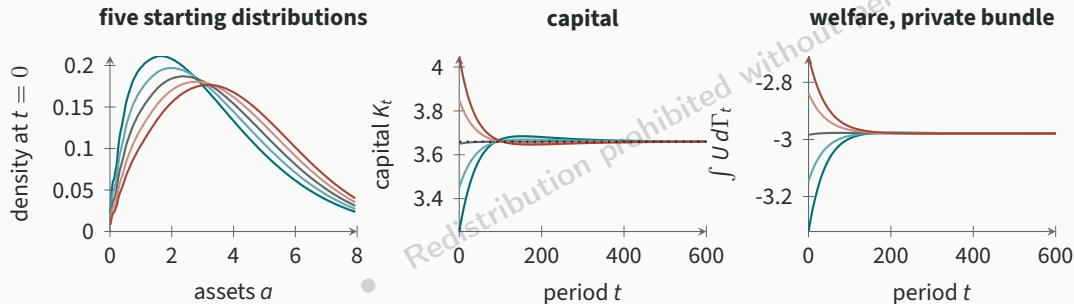
Same Capital, Different Inequality



- ▶ All start at $K_0 = 3.660$, Gini 0.301–0.372; all reach Gini 0.341, from below and above.
- ▶ **K leaves and returns (3.620–3.723). A state summarised by its mean would see nothing.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B. Dotted: Stage 1A's K^* .

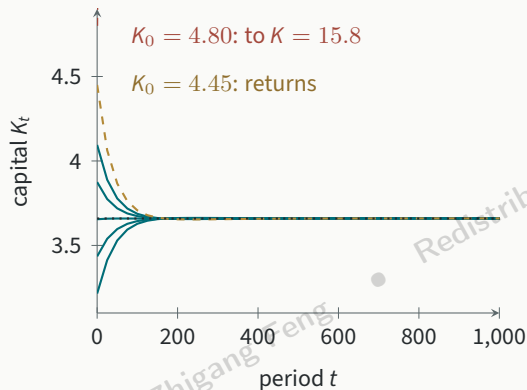
Same Spread, Different Capital



- ▶ K_0 from 3.25 to 4.05: K returns in about a hundred periods, the Gini far later.
- ▶ **Welfare along the path, from any start: what Stage 1A cannot return.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B. Dotted: Stage 1A's K^* . No government, so no public good.

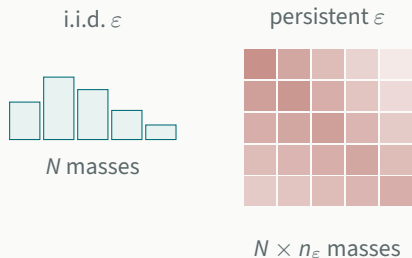
One Attractor, and the Edge of the Band



- ▶ 5 starts with K_0 from 3.21 to 4.09: one Γ^* , terminal K within 3.4×10^{-8} .
- ▶ Within 10^{-3} after 50–475 periods, 10^{-6} after 1125–1325: βR near one makes it slow.
- ▶ Outside the band, $K_0 = 4.45$ returns; $K_0 = 4.80$ runs to the top of the grid.
- ▶ **The rule is valid where it was trained.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B; dotted: Stage 1A's K^* .

Where the Characterisation Gets Hard



a and ε become correlated: the poor are poor because their ε was low

- Persistent ε : $\Gamma(\varepsilon, a)$ no longer factorises. The state is a joint histogram, and $\hat{\Phi}$ must carry the chain Π .
- With i.i.d. risk only 0.009% sit at the limit. In the lab's economy, with persistence 0.9, 2.9% do: runs of low ε walk them down to it.
- An aggregate shock z makes the map $\Phi(\Gamma, z, z')$ random, to be learned on every branch.
- **Each step multiplies the size of $\hat{\Phi}$ and of every rollout in memory.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the lab, block A; Krusell & Smith, *JPE* (1998).

Bridge: When Φ Cannot Be Written Down, Simulate It

This deck. Γ is a histogram, Φ a matrix computed in the graph: exact, deterministic, differentiable.

persistence, aggregate risk

Next. Γ is a simulated panel, and the network learns from the states that the panel visits.

- ▶ Deep Equilibrium Nets and DeepHAM simulate households, and train where the simulation goes.
- ▶ The price: sampling noise, no derivative through a head count, coverage only where the policy goes.
- ▶ The gain: persistence and aggregate risk cost a longer simulation, not a bigger matrix.
- ▶ **Deck 4.2: three such methods. Deck 4.4: this histogram, with a government.**

Source: Azinovic, Gaegauf & Scheidegger, *IER* (2022); Han, Yang & E (2026).